Optimal Pricing Design for Coordinated and Uncoordinated IoT Networks

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Abstract—An Internet of Things (IoT) system can include several different types of service providers, who sell IoT service, network service, and computation service to customers, either jointly or separately. A deep understanding of complicated coupling among these providers in terms of pricing and service decisions is critical to the success of IoT networks. This paper studies the impact of the provider interaction structures on the overall IoT system with heterogeneous customers. Specifically, we first study a generic IoT scenario with three interaction structures: coordinated, vertically-uncoordinated, and horizontally-uncoordinated structures. Despite the challenging non-convex optimization problems involved in modeling and analyzing these structures, we successfully obtain the closed-form optimal pricing strategies of providers in each interaction structure. We further extend the analysis to a specific IoT scenario with local computation capability (e.g., Internet of Vehicles (IoV)). We prove that the coordinated structure is better than two uncoordinated structures for both providers and customers, as it avoids selfish price markup behaviors in uncoordinated structures. Between the two uncoordinated structures, when customers' demand variance is large and utility-cost ratio is medium, vertically-uncoordinated structure is better than horizontal one for both providers and customers, due to the complementary providers' competition in horizontally-uncoordinated structure. Counter-intuitively, we identify that providers' optimal prices do not change with their costs at the critical point of customers' full participation in the vertically-uncoordinated structure.

Index Terms—Heterogeneous customers, interaction structure comparison, Internet of Things, pricing

1 Introduction

1.1 Background and Motivations

With the fast development of machine-type communications, the Internet of Things (IoT) becomes increasingly popular in many fields such as healthcare, transportation, and manufacturing [2]. The next generation (5 G/6 G) mobile networks further enrich the application scenarios of IoT systems by providing the capability of handling massive connectivity and achieving ultra-reliable low-latency communications.

To fully exploit the potential of IoT, we need to solve both technical challenges (e.g., the development and application of key enabling technologies [3]) and economic challenges (e.g., pricing and business models [4]). As the prerequisite for IoT system deployment, the technical challenges have been extensively studied in literature (e.g., [5], [6]). However, economic issues still need further exploration to promote the profitable and sustainable development of IoT applications [7]. Computing Technology Industry Association shows that global IoT spending is predicted to reach $520 billion in 2021 [8], but only 23% of IoT channel companies made money from their IoT business in 2016 [9]. Therefore, it is essential to design an appropriate pricing mechanism for IoT systems to help fully realize the benefits of technology advances.

There are several key challenges to tackle to reach the above goal. First, service providers need to serve massive number of IoT customers, who have heterogeneous demands on computation and network resources as well as different usage levels of the IoT service. It is challenging for service providers to take all customers' demands and strategies into consideration for designing the optimal prices.

Second, there exist multiple kinds of service providers (e, g., IoT service provider (IoTSP), network service provider (NSP), and cloud service provider (CSP)) with complementary services but diverse objectives in IoT systems. The complementary services encourage them to jointly improve customers' experiences, but their selfish profit maximization objectives motivate them to compete with each other. Hence, providers are engaging in a complicated competition-cooperation relationship.

Third, the complex interaction among customers and service providers also significantly influences the price design. Currently, there exist three interaction structures in IoT systems, each of which requires different pricing considerations.

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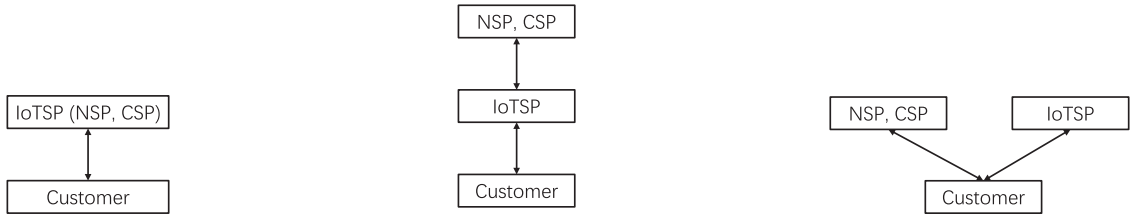

(a) Coordinated interaction structure. (b) Vertically-uncoordinated interaction structure. (c) Horizontally-uncoordinated interaction structure.

Fig. 1. Interaction structures.

1) *Coordinated interaction structure*. As shown in Fig. 1a, IoTSP (e.g., Amazon's AWS IoT platform [10]) also serves as NSP and CSP. It provides integrated services to customers, including IoT service and the corresponding network and computation services.[1]
2) *Vertically-uncoordinated interaction structure*. As shown in Fig. 1b, IoTSP (e.g., BYD Auto [11]) procures network and computing resources from NSP (e.g., China Mobile [12]) and CSP (e.g., Huawei Cloud [13]), respectively, and then provides services for customers. Customers only need to pay IoTSP for the IoT service (e.g., automatic driving related services), and they do not interact with NSP and CSP directly.
3) *Horizontally-uncoordinated interaction structure*. As shown in Fig. 1c, customers purchase IoT service from IoTSP (e.g., smartwatch from Cosmo [14]) and supporting resources (e.g., SIM card and cloud computation/storage) from NSP and CSP, respectively. Service providers do not interact with each other directly.

The above considerations motivate us to answer the first key question in this paper:

Question 1. *What are the service providers' optimal prices for serving heterogeneous customers under different interaction structures in IoT systems? (Sections 2, 3, and 4)*

The performance comparison of interaction structures is also of substantial significance. Different interaction structures will lead to different service providers' optimal prices, profits, market shares, as well as customers' payoffs. When there is a free market where the above interaction structures coexist, studying the superiority of each interaction structure will facilitate the structure selection of every service provider and customer. Structure comparison also provides insights into the policy design of the market regulator. This will lead to the second key question of this paper:

Question 2. *Which interaction structure is the best in terms of service providers' profits, market share, or customers' payoffs? (Section 5)*

We will analyze the above two questions in a generic IoT scenario systematically. Furthermore, we can extend our analysis to different specific IoT scenarios with unique characteristics and challenges. As a representative example, we consider the rapidly growing Internet of Vehicles (IoV) scenario, where each customer device (i.e., vehicle) has considerable computation capability. A vehicle can have the power of about 20 personal computers and process 25 GB of data every hour, with some of that data sent back to the car manufacturer or cloud in real-time for processing [15]. Also, with the fast development of hardware technology, some smartwatches and smartphones also have adequate computing power. Thus, this motivates us to show that we can easily extend our generic model and analysis by considering customers' local computation and to answer the following key question:

Question 3. *How does customers' (partial) local computation affect service providers' optimal prices and profit as well as customers' optimal decisions and payoffs? (Section 6)*

1.2 Contributions

We summarize our key novelty and contributions below.

- *Novel IoT Model with Multiple Interaction Structures:*To the best of our knowledge, this is the first analytical work that studies the optimal pricing for heterogeneous IoT customers under different interaction structures. Such a study is practically important for the sustainable development of profitable IoT systems.
- *Optimal Pricing Design under Different Interaction Structures:* Given heterogeneous customers, service providers' objectives are challenging piece-wise functions with non-convex pieces. We obtain the global optimal solutions to the pricing problems in different interaction structures by properly decomposing them into several subproblems, performing transformations of variables and functions, and analyzing the properties of intermediate implicit functions.
- *Comparison of Interaction Structures under Different Metrics:*We prove that (i) full coordination among service providers is better than uncoordination for both providers and customers, as it avoids price markup behaviors in the uncoordinated structures; (ii) between two uncoordinated structures, when customers' demand variance is large and utility-cost ratio is medium, vertically-uncoordinated structure is better than horizontally-uncoordinated structure for both service providers and customers. This is because the competition among complementary service providers in the horizontally-uncoordinated structure increases their prices when customers' demand is inelastic.
- *Impact of Partial Local Computation:* We further extend the model by jointly considering local computation and cloud computation. We prove that a larger local

1. Note that this structure can also be regarded as three service providers cooperating with each other to jointly maximize their total profit.

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or cloud computation cost will decrease both service providers' profit and customers' payoffs, but the impact on optimal prices is not monotonic as customers have different demand elasticity at different costs. The impact of local computation fraction depends on the comparison between local and cloud computation costs of all participants.

- *Performance Evaluation and Insights:* We identify a counter-intuitive *price frozen effect* in the vertically-uncoordinated structure, where the optimal prices do not change in the costs. This corresponds to the critical point of customers' full participation, where service providers' price rise will lead to profit reduction due to customers' demand drop. Moreover, we numerically show that compared with the state-of-the-art benchmarks, our proposed pricing mechanisms can increase service providers' profits by up to 135.48%.

1.3 Related Work

Studies on IoT pricing have sprung up in recent years, including some game-theory-based analysis (e.g., [16], [17]). However, most economic literature about IoT systems focused on the optimization issue of one (type of) service provider. For example, some work only studied NSP's optimal pricing and network resource allocation scheme (e.g., [18], [19], [20], [21], [22]), some just focused on CSP's optimal pricing and the optimization of computation task offloading (e.g., [23], [24], [25]), and the others only investigated IoTSP's optimal pricing and data collection incentives (e.g., [26], [27], [28], [29]). These prior studies did not consider the impact of interaction structures among different types of service providers, which are challenging to analyze yet practically significant. Also, jointly considering all these service providers in a single market will lead to fundamental changes to each service provider's pricing strategy, which requires a new holistic modeling framework.

The seminal papers by Ghosh and Sarkar in [30] and Lee in [16] investigated the pricing strategies of different service providers in IoT systems, with several limitations. The authors in [30] focused on homogeneous customers, and they did not consider service providers' resource/operation costs or customers' optimal strategies. They assumed that customers have the same and fixed resource requirement, ignoring the strategic demands of different customers in IoT systems. Moreover, the paper [16] considered that each customer only decides whether to participate based on a reservation price but will not change his service usage level according to different prices and costs (if he participates). It will be more comprehensive to further consider the heterogeneous network/computing resource demands of customers, which can significantly affect service providers' costs and pricing.

Building upon these earlier works, we consider a more general and practical model of customers and service providers in IoT systems, provide a more comprehensive pricing design with closed-form solutions, and investigate the effect of interaction structures.

The rest of the paper is organized as follows. The system model is described in Section 2. We study the optimal pricing of the service providers in the coordinated structure and uncoordinated structures in Sections 3 and 4, respectively. The comparison of different interaction structures is presented in Section 5. Besides the analysis of generic IoT systems, we further extend the model by considering some special characteristics of partial local computation in Section 6. We perform simulations in Section 7 and conclude in Section 8.

2 System Model

We consider a generic IoT system (e.g., a system of smart wearable devices[2]) consisting of a network service provider (NSP), a cloud service provider (CSP), an IoT service provider (IoTSP), and N customers. Some of these providers may be controlled by the same decision maker. In Sections 2.1 and 2.2, we will specify their different strategies and payoffs in each interaction structure. In Section 2.3, we will formulate their strategic interactions using game-theoretical models.

2.1 Strategies of Customers and Service Providers

2.1.1 Heterogeneous Customers' Strategies

Consider a set $\mathcal{N} = \{1, 2, \ldots, N\}$ of customers (e.g., people who use automatic driving services) in a practical IoT system where the number of customers is large, i.e., $N \to \infty$ [32], [33].

The customers are distinguished by their computation and network resource needs for using the IoT service. Each time customer n uses the IoT service, there exists some computation load (in bits) that he cannot process locally (e.g., large-scale model training).[3] We denote the required cloud computation resource (in CPU cycles) for processing customer n's computation load as $c_n > 0$, and we denote the required network resource (in bits) for transferring the computation load as $d_n > 0$.[4]

Following a typical model in the literature (e.g., [34], [35]), we assume $c_n = \xi_n d_n$, where ξ_n is a constant coefficient related to communication and computation capabilities. We assume that customers in the same IoT system use the same type of IoT devices and technologies, hence $\xi_n = \xi, \forall n \in \mathcal{N}$. In this case, customers' heterogeneity can be characterized by the network demand d, and we assume that the probability density function of customers' distribution is $g(d)$ on support $[d^{\min}, d^{\max}]$.[5]

Given the resource demands, each customer $n \in \mathcal{N}$ needs to decide his IoT service usage level $\ell_n \geq 0$ (e.g., number of usages per month) based on service providers' prices.

2. In such a system, customers want to connect their smart wearable devices (e.g., smartwatches) to their phones, transmit data (e.g., real-time heart rate) to the phone (and in some cases further to an online server), and analyze the data to extract useful information (e.g., health assessment, disease diagnosis, and professional medical treatment) [31]. In this case, the IoTSP provides IoT services such as device connection, NSP provides network services for data transmission, and CSP helps with the data analysis and information extraction.

3. Here, we consider that a customer's cost of local background computation is a constant, which will not affect his strategy optimization. We will extend our model by considering both local and cloud computation in Section 6.

4. For example, the communication cost needed for sending the data to the cloud and receiving the computation result from the server.

5. Note that each customer has a different computation demand from others, which is a constant known by himself but not by service providers. The service provider knows the distribution of all customers' demands.

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2.1.2 Service Providers' Strategies

Each service provider decides the price for its service. We use p_N, p_C, and p_I to denote the unit prices of NSP's network resource, CSP's computation resource, and IoTSP's IoT service,[6] respectively.

2.2 Payoffs of Customers and Service Providers

2.2.1 Customers' Payoffs

A customer's payoff is the difference between his utility and the payment to service providers. We denote customer n's utility of using the IoT service as

$$f(\ell_n) \triangleq A\log(1+\ell_n), \tag{1}$$

where the logarithmic function captures the widely considered diminishing marginal return feature of the utility function [36], [37]. Intuitively, each customer's utility monotonically increases in his usage level, as he enjoys more services. However, when there is already an adequate usage level, further increasing the usage will only bring minor gain to his utility. We assume that customers have the same utility coefficient $A > 0$.[7]

A customer's payment to service providers is different in three interaction structures:

- In coordinated and vertically-uncoordinated structures, customers only pay IoTSP. Thus, the payment of customer n is the product of IoTSP's unit price and the customer's demand, i.e., $p_I d_n \ell_n$. Customer n's payoff is

$$U_n = \quad A\log(1+\ell_n) - p_I d_n \ell_n. \tag{2}$$

- In the horizontally-uncoordinated structure, customers need to pay all three service providers directly. Therefore, the payment of customer n is the sum of three parts, i.e., $(p_I + p_N + p_C\xi)d_n\ell_n$. Customer n's payoff is

$$U_n = \quad A\log(1+\ell_n) - (p_I + p_N + p_C\xi)d_n\ell_n. \tag{3}$$

2.2.2 Service Providers' Profits

We use $C_N > 0$, $C_C > 0$, and $C_I > 0$ to denote the NSP's cost per unit of network resource, CSP's cost per unit of computation resource, and IoTSP's operation cost per unit of network resource consumption, respectively[8].

Service providers' profits depend on interaction structures:

- *Coordinated structure*. The profit of IoTSP (which also serves as NSP and CSP) is the difference of the revenue from customers and the costs:

$$W_I = \int_{d^{\min}}^{d^{\max}} \ell d(p_I - C_N - C_C\xi - C_I)g(d)\mathrm{d}d, \tag{4}$$

where the costs consist of the network cost, the computation cost, and the operation cost.

- *Vertically-uncoordinated structure*. The NSP's profit is the difference of the revenue and the network resource cost:

$$W_N = \int_{d^{\min}}^{d^{\max}} \ell d(p_N - C_N)g(d)\mathrm{d}d. \tag{5}$$

The CSP's profit is the difference of its revenue and the computation resource cost:

$$W_C = \int_{d^{\min}}^{d^{\max}} \ell\xi d(p_C - C_C)g(d)\mathrm{d}d. \tag{6}$$

The IoTSP's profit is the difference of the revenue from customers and its costs, including the operation cost and the payment to NSP and CSP, i.e.,

$$W_I = \int_{d^{\min}}^{d^{\max}} \ell d(p_I - p_N - p_C\xi - C_I)g(d)\mathrm{d}d. \tag{7}$$

- *Horizontally-uncoordinated structure*. The profits of NSP and CSP are the same as (5) and (6), respectively. The IoTSP's profit in this case is the difference of the revenue from customers and the operation cost.

$$W_I = \int_{d^{\min}}^{d^{\max}} \ell d(p_I - C_I)g(d)\mathrm{d}d. \tag{8}$$

2.3 Game-Theoretic Models

We formulate the strategic interaction of customers and providers as a multi-stage game in each interaction structure.

- In coordinated and horizontally-uncoordinated structures, we use two-stage Stackelberg games to model the interactions. Each service provider announces its price in Stage I, and then customers decide their usage levels in Stage II. The difference is that in the horizontally-uncoordinated structure, service providers participate in a non-cooperative game in Stage I, due to their simultaneous decision making and diverse objectives.
- In the vertically-uncoordinated structure, we use a three-stage game to model the interactions. NSP and CSP first announce their resource prices in Stage I, then IoTSP determines its price for the IoT service in Stage II, and finally customers decide their usage levels in Stage III.

Next, we will use backward induction to derive the optimal strategies of customers and service providers in each interaction structure. For ease of reading, we list the key notations in Table 1. Moreover, to differentiate the results in three interaction structures, we use superscripts co, vu, and hu to represent coordinated, vertically-uncoordinated, and horizontally-uncoordinated structures, respectively.

3 Optimal Joint Pricing in Coordinated Structure

In this section, we analyze the optimal prices of service providers in the coordinated structure. Based on the backward

6. Since customers' resource demands can be uniquely characterized by the network demand, the unit of IoTSP's price p_I is also per network consumption.

7. Our analysis approach can be easily applied to the case where customers have different utility coefficients, and we will show through analysis and simulation in Section 7.4 that the key insights still remain valid under different utility coefficients.

8. Such a linear resource/operation cost model has been widely used in literature (e.g., [38], [39]).

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TABLE 1
Key Notations

d_n	Required network resource for each usage of IoT service of customer n
$d^{\min}$	Customers' minimum required network resource for each usage of IoT service
$d^{\max}$	Customers' maximum required network resource for each usage of IoT service
c_n	Required computation resource for each usage of IoT service of customer n
ℓ_n	IoT service usage level of customer n
A	Customers' utility coefficient
p_N	Unit price of NSP's network resource
p_C	Unit price of CSP's computation resource
p_I	Unite price of IoTSP's IoT service
C_N	NSP's unit network cost
C_C	CSP's unit computation cost
C_I	IoTSP's unit operation cost
C	Total unit cost $C \triangleq C_N + C_C\xi + C_I$

induction, we first derive customers' optimal strategies and then analyze IoTSP's optimal price.

3.1 Stage II: Customers' Optimal Decisions

Given IoTSP's price p_I^{co} in Stages I, each customer n decides his optimal IoT service usage level ℓ_n^{co} to maximize his payoff in Stage II by solving the following problem:

Problem 1 (Customer n's Optimization Problem).

$$\max \quad A\log(1+\ell_n^{co}) - p_I^{co} d_n \ell_n^{co} \tag{9a}$$
$$\text{var.} \quad \ell_n^{co} \in [0,\infty). \tag{9b}$$

As Problem 1 is a convex optimization problem, we can derive the closed form of the optimal solution in Lemma 1.

Lemma 1. *A customer n's optimal IoT service usage level in the coordinated structure is*

$$\ell_n^{co^*} = \begin{cases} \frac{A}{p_I^{co} d_n} - 1, & \text{if } p_I^{co} \le \frac{A}{d_n}, \quad (10a) \\ 0, & \text{if } p_I^{co} > \frac{A}{d_n}. \quad (10b) \end{cases}$$

Proof of Lemma 1 is given in Appendix 1 (available online). According to Lemma 1, the optimal IoT service usage level $\ell_n^{co^*}$ decreases in IoTSP's price (i.e., p_I^{co}) and the customer's resource demand for each usage (i.e., d_n). As shown in (10b), a customers will not use the IoT service if the service is too expensive or the resource consumption of each service usage is too large (e.g., when facing a complex road condition in the vehicle network).

3.2 Stage I: IoTSP's Optimal Price

Considering the optimal strategies of customers in Stage II, IoTSP decides the optimal price p_I^{co} to maximize its profit in Stage I.

To present the closed-form solution, we consider a widely-adopted uniform distribution on $[d^{\min}, d^{\max}]$ in this paper [40], [41], [42][9], i.e.,

9. Our analysis approach can be easily applied to other distributions. We will show through analysis and simulation in Section 7.4 that the key insights still remain valid under different demand distribution.

Assumption 1. *Customers' network resource demand d follows a uniform distribution with the probability density function:*

$$g(d) \triangleq \begin{cases} \frac{1}{d^{\max}-d^{\min}}, & \text{if } d^{\min} \le d \le d^{\max}, \\ 0, & if\ d < d^{\min} \text{ or } d > d^{\max}. \end{cases} \tag{11}$$

In this case, the IoTSP's optimization problem is

Problem 2 (IoTSP's Optimization Problem).

$$\max \frac{1}{d^{\max}-d^{\min}} \int_{d^{\min}}^{d^{\max}} \ell^{co} d[p_I^{co} - (C_N + C_C\xi + C_I)]\mathrm{d}d \tag{12a}$$
$$\text{var. } p_I^{co} \in [0,\infty). \tag{12b}$$

Based on Lemma 1, we can show that IoTSP's objective (12a) is a piece-wise function with non-convex pieces. Due to the uncertain numerical relationship of parameters, the convexity and monotonicity of each piece have multiple possibilities. Hence, it is hard to solve this problem by simply decomposing each piece into convex subsegments. To address this issue, we properly divide the analysis into several cases depending on the system parameters, and for each case we further decompose the problem into several subproblems. The subproblems are not necessarily convex but sufficient for us to obtain the closed-form solutions through monotonicity analysis.

We define

$$C \triangleq C_N + C_C\xi + C_I \tag{13}$$

as the total unit cost, and the optimal price of IoTSP is as follows.

Theorem 1. *The IoTSP's optimal price in the coordinated structure is*

$$p_I^{co^*} = \begin{cases} \sqrt{\frac{2AC}{d^{\max}+d^{\min}}}, & \text{if } C \le \frac{d^{\max}+d^{\min}}{2\,d^{\max^2}} A, \\ \frac{\sqrt{A^2+8ACd^{\min}}-A}{2\,d^{\min}}, & \text{if } \frac{d^{\max}+d^{\min}}{2\,d^{\max^2}} A < C < \frac{A}{d^{\min}}, \\ \frac{A}{d^{\min}}, & \text{if } C \ge \frac{A}{d^{\min}}. \end{cases} \tag{14}$$

Proof of Theorem 1 is given in Appendix 2, available in the online supplemental material. According to Theorem 1, when the unit cost C increases, IoTSP's optimal price will also increase. When C is too large (i.e., $C \ge A/d^{\min}$), IoTSP will set a high enough price such that no customer will use its service. However, as we will see in next section, service providers in uncoordinated structures will have more complex and sometimes counter-intuitive behaviors, e.g., they may not raise their prices when their costs increase.

4 Optimal Pricing in Vertically and Horizontally Uncoordinated Structures

In this section, we analyze service providers' optimal prices in vertically-uncoordinated and horizontally-uncoordinated structures, considering the optimal strategies of customers.

4.1 Vertically-Uncoordinated Interaction Structure

In the three-stage game of the vertically-uncoordinated structure, NSP and CSP first decide their resource prices in Stage I,

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$$p_N^{vu^*} + p_C^{vu^*}\xi = \begin{cases} \frac{1}{6(d^{\max}+d^{\min})}\left(A - y_1^{\frac{1}{3}} - y_2^{\frac{1}{3}}\right) - C_I, & \text{if } C \le \frac{Ad^{\min}(d^{\max}+d^{\min})}{2\ d^{\max^3}}, \quad (19a) \\ \frac{d^{\max}+d^{\min}}{2\ d^{\max^2}}A - C_I, \text{if} & \frac{Ad^{\min}(d^{\max}+d^{\min})}{2\ d^{\max^3}} < C \le \frac{A(d^{\max^2}+d^{\max}d^{\min}+2\ d^{\min^2})}{4\ d^{\max^3}}, \quad (19b) \\ \frac{2\ A^3}{9\ d^{\min^3}}\left(y_3^{\frac{1}{3}} + y_4^{\frac{1}{3}} - \frac{d^{\min}}{4\ A}\right)\left(y_3^{\frac{1}{3}} + y_4^{\frac{1}{3}} + \frac{5\ d^{\min}}{4\ A}\right) - C_I, & \text{if } \frac{A(d^{\max^2}+d^{\max}d^{\min}+2\ d^{\min^2})}{4\ d^{\max^3}} < C \le \frac{A}{d^{\min}}, \\ \frac{A}{d^{\min}} - C_I, & \text{if } C > \frac{A}{d^{\min}}, \quad (19c) \end{cases}$$

where

$$\begin{aligned} y_1 &= -A^3 - 18\ A^2C(d^{\max}+d^{\min}) - 54(d^{\max}+d^{\min})^2AC^2 \\ &\quad + 6AC(d^{\max}+d^{\min})\sqrt{81(d^{\max}+d^{\min})^2\ C^2 + 6(d^{\max}+d^{\min})AC}, \\ y_2 &= -A^3 - 18\ A^2C(d^{\max}+d^{\min}) - 54(d^{\max}+d^{\min})^2AC^2 \\ &\quad - 6AC(d^{\max}+d^{\min})\sqrt{81(d^{\max}+d^{\min})^2\ C^2 + 6(d^{\max}+d^{\min})AC}, \\ y_3 &= \frac{d^{\min^3}(A+27Cd^{\min})}{8\ A^4} + \frac{3\ d^{\min^3}}{4\ A^2}\sqrt{\left(\frac{1}{16}+\frac{9Cd^{\min}}{2\ A^2}\right)^2 + \frac{5}{16\ A^2}\left(\frac{1}{4}+\frac{3Cd^{\min}}{A}\right)}, \\ y_4 &= \frac{d^{\min^3}(A+27Cd^{\min})}{8\ A^4} - \frac{3\ d^{\min^3}}{4\ A^2}\sqrt{\left(\frac{1}{16}+\frac{9Cd^{\min}}{2\ A^2}\right)^2 + \frac{5}{16\ A^2}\left(\frac{1}{4}+\frac{3Cd^{\min}}{A}\right)}. \end{aligned}$$

then IoTSP determines its IoT service price in Stage II, and finally customers decide their usage levels in Stage III. Next we use backward induction to derive their optimal strategies.

4.1.1 Stage III: Customers' Optimal Decisions

Each customer n's optimization problem is the same as Problem 1 in the coordinated structure, leading to the same optimal strategies in Lemma 1.

4.1.2 Stage II: IoTSP's Optimal Price

IoTSP's optimization problem in this case is as follows.

Problem 3 (IoTSP's Optimization Problem).

$$\max \frac{1}{d^{\max}-d^{\min}}\int_{d^{\min}}^{d^{\max}} \ell^{vu} d[p_I^{vu} - (p_N^{vu} + p_C^{vu}\xi + C_I)]\mathrm{d}d \quad (15a)$$

$$\text{var. } p_I^{vu} \in [0,\infty). \quad (15b)$$

We define

$$P \triangleq p_N^{vu} + p_C^{vu}\xi + C_I \quad (16)$$

as IoTSP's the unit cost in the vertically-uncoordinated structure. Then, IoTSP's optimal price is as follows.

Theorem 2. *The IoTSP's optimal price in the vertically-uncoordinated structure is*

$$p_I^{vu^*} = \begin{cases} \sqrt{\frac{2AP}{d^{\max}+d^{\min}}}, & \text{if } 0 \le P \le \frac{d^{\max}+d^{\min}}{2\ d^{\max^2}}A, \\ \frac{\sqrt{A^2+8APd^{\min}}-A}{2\ d^{\min}}, & \text{if } \frac{d^{\max}+d^{\min}}{2\ d^{\max^2}}A < P < \frac{A}{d^{\min}}, \\ \frac{A}{d^{\min}}, & \text{if } P \ge \frac{A}{d^{\min}}. \end{cases} \quad (17)$$

Proof of Theorem 2 is given in Appendix 3, available in the online supplemental material. Compared with the optimal price in the coordinated structure, the only difference is IoTSP's cost, which is C in Theorem 1 and P in Theorem 2. It is clear that in the vertically-uncoordinated structure, the IoTSP's optimal price still increases in its cost. However, next we will show the counter-intuitive behaviors that the optimal prices of NSP and CSP do not always monotonically increase with their costs.

4.1.3 Stage I: NSP's and CSP's Optimal Prices

To highlight the difference of strategies between IoTSP and resource service providers (i.e., NSP and CSP), we consider NSP and CSP jointly maximize their total profit.[10] The joint optimization problem of NSP and CSP is as follows.

Problem 4 (Optimization Problem of NSP and CSP).

$$\max \frac{1}{d^{\max}-d^{\min}}\int_{d^{\min}}^{d^{\max}} \ell^{vu} d(p_N^{vu} + p_C^{vu}\xi - C_N - C_C\xi)\mathrm{d}d \quad (18a)$$

$$\text{var. } p_N^{vu} \in [0,\infty), p_C^{vu} \in [0,\infty). \quad (18b)$$

After substituting the optimal strategies of customers and IoTSP in Stages II and III into the objective function (18a), we end up with a piece-wise non-convex optimization problem. It is much more complicated and challenging to analyze than Problem 2, due to the three-stage interactions here. To tackle this challenge, we leverage proper transformations of variables and functions, and fully exploit some key properties (e.g., monotonicity and maximum values) of the transformed functions and intermediate implicit functions. We present the closed-form global optimal solution in Theorem 3.

10. There also exists another case where the NSP and CSP optimize their objectives respectively. The impact on the IoTSP's strategy will be similar in this case, but the analysis will be more complicated. Due to space limit, we will leave it for future work.

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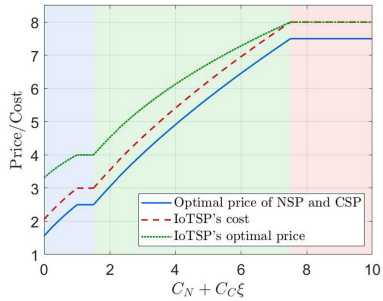

Fig. 2. Service providers' optimal prices versus the cost of NSP and CSP in the vertically-uncoordinated structure. Three shadow regions: blue means full customer participation, green means partial participation, and pink means no participation.

Theorem 3. *The optimal prices of NSP and CSP in the vertically-uncoordinated structure are summarized in (19).*[11]

Proof of Theorem 3 is given in Appendix 4, available in the online supplemental material. We use Figs. 2, 3, and 4 to illustrate the interesting results and insights in Theorem 3.

Fig. 2 shows a *price frozen effect* in $C_N + C_C\xi \in [1, 1.5]$ (corresponding to (19b)), where the optimal prices of NSP and CSP do not change with their costs. In this case, IoTSP will have a fixed payment to NSP and CSP and thus set a fixed price for customers.

Fig. 3 shows that this effect is because the unchanged price of NSP and CSP corresponds to the critical point of customers' full participation. To be specific, if the price $p_N^{vu} + p_C^{vu}\xi$ is larger than the optimal price $p_N^{vu^*} + p_C^{vu^*}\xi$ in (19b), some customers will not participate, which result in the profit reduction of NSP and CSP; If $p_N^{vu} + p_C^{vu}\xi$ is smaller than $p_N^{vu^*} + p_C^{vu*}\xi$, unit profit decreases but customers' total demand cannot significantly increase, as all customers have already participated.

As illustrated in Fig. 4, we present the profits of NSP and CSP when $C_N + C_C\xi$ takes four values $v_1 < v_2 < v_3 < v_4$, respectively. When the cost increases from v_2 to v_3, the profit of NSP and CSP decreases, but the optimal price that maximizes the profit remains at 2.5.

Next we will study the horizontally-uncoordinated structure. We will show that service providers' simultaneous decision-making will lead to a combination of sequential and parallel non-cooperative games among service providers and customers, resulting in completely different pricing strategies.

4.2 Horizontally-Uncoordinated Interaction Structure

In this structure, customers purchase services from all service providers directly. We will first analyze customers' optimal strategies in Stage II, and then we will study service providers' non-cooperative game in Stage I.

4.2.1 Stage II: Customers' Optimal Decisions

Given the prices p_I^{hu}, p_N^{hu}, and p_C^{hu} in Stages I, each customer n decides his optimal IoT service usage level ℓ_n^{hu} to maximize his payoff in Stage II, as summarized in Problem 5.

11. The specific values of $p_N^{vu^*}$ and $p_C^{vu^*}$ can be chosen arbitrarily, as long as their weighted sum equals the value in (19) and they are larger than the respective costs (i.e., C_N and C_C).

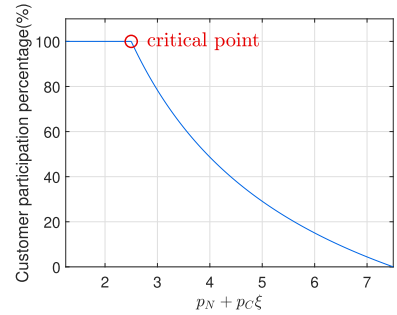

Fig. 3. Customers' participation percentage versus the price of NSP and CSP in the vertically-uncoordinated structure.

Problem 5 (Customer n's Optimization Problem).

$$\max \quad A\log(1+\ell_n^{hu}) - (p_I^{hu} + p_N^{hu} + p_C^{hu}\xi)d_n\ell_n \tag{20a}$$

$$\text{var.} \quad \ell_n^{hu} \in [0, \infty). \tag{20b}$$

Lemma 2 summarizes the optimal solution of Problem 5, which is customers' optimal strategies.

Lemma 2. *A customer n's optimal IoT service usage level in the horizontally-uncoordinated structure is*

$$\ell_n^{hu^*} = \begin{cases} \frac{A}{(p_I^{hu}+p_N^{hu}+p_C^{hu}\xi)d_n} - 1, & \text{if } p_I^{hu} + p_N^{hu} + p_C^{hu}\xi \le \frac{A}{d_n}, \\ 0, & \text{if } p_I^{hu} + p_N^{hu} + p_C^{hu}\xi > \frac{A}{d_n}. \end{cases}$$

Proof of Lemma 2 is given in Appendix 5, available in the online supplemental material. Lemma 2 shows that the optimal IoT service usage level ℓ_n^* decreases in his resource demand and the total price.

4.2.2 Stage I: Optimal Prices of IoTSP, NSP, and CSP

In Stage I, three service providers engage in a non-cooperative game, where the NSP and CSP jointly act as one player. We will demonstrate the surprising result that such a non-cooperative competition can make service providers simultaneously increase their prices.

We formally define service providers' non-cooperative pricing game and the corresponding equilibrium as follows.

Definition 1 (Stage I: Service Providers' Pricing Game).

- *Players: IoTSP and NSP&CSP.*
- *Strategy space: IoTSP chooses its price $p_I^{hu} \in [0, \infty)$; NSP and CSP jointly choose their prices $p_N^{hu} \in [0, \infty)$ and $p_C^{hu} \in [0, \infty)$.*

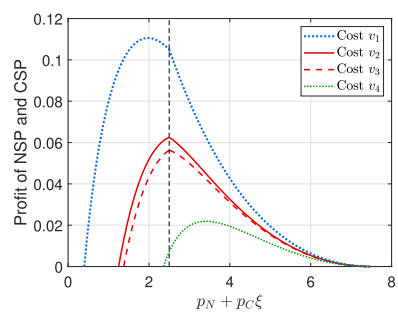

Fig. 4. Profit of NSP and CSP under different prices when their cost $C_N + C_C\xi$ takes four values $v_1 < v_2 < v_3 < v_4$ in the vertically-uncoordinated structure.

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$$\begin{cases} \text{If } C \le \frac{d^{\min}}{d^{\max 2}} A, & \text{then } p_I^{hu^*} = \frac{A-2(d^{\max}+d^{\min})(C_N+C_C\xi-C_I)+\sqrt{A^2+4AC(d^{\max}+d^{\min})}}{4(d^{\max}+d^{\min})}, \quad (23a) \\ & p_N^{hu^*} + p_C^{hu^*}\xi = \frac{A+2(d^{\max}+d^{\min})(C_N+C_C\xi-C_I)+\sqrt{A^2+4AC(d^{\max}+d^{\min})}}{4(d^{\max}+d^{\min})}, \quad (23b) \\ \text{If } \frac{d^{\min}}{d^{\max 2}} A < C < \frac{A}{d^{\min}}, & \text{then } p_I^{hu^*} = \frac{1}{2}\left(\sqrt{\frac{AC}{d^{\min}}} - C_N - C_C\xi + C_I\right), p_N^{hu^*} + p_C^{hu^*}\xi = \frac{1}{2}\left(\sqrt{\frac{AC}{d^{\min}}} + C_N + C_C\xi - C_I\right), \\ \text{If } C \ge \frac{A}{d^{\min}}, & \text{then } p_I^{hu^*} = C_I,\ p_N^{hu^*} + p_C^{hu^*}\xi = \frac{A}{d^{\min}} - C_I. \quad (23c) \end{cases}$$

- *Payoff function: IoTSP maximizes its profit*

$$W_I(p_I^{hu}; p_N^{hu}, p_C^{hu}) = \frac{1}{d^{\max} - d^{\min}} \int_{d^{\min}}^{d^{\max}} \ell^{hu} d(p_I^{hu} - C_I) \mathrm{d}d; \quad (21)$$

NSP and CSP maximize their total profit

$$W_{NC}(p_N^{hu}, p_C^{hu}; p_I^{hu}) = \frac{1}{d^{\max} - d^{\min}} \int_{d^{\min}}^{d^{\max}} \ell^{hu} d(p_N^{hu} + p_C^{hu}\xi - C_N - C_C\xi) \mathrm{d}d. \quad (22)$$

Definition 2 (Pricing Equilibrium in Stage I). *The equilibrium of Service Providers' Pricing Game is a pricing profile $(p_I^{hu^*}, p_N^{hu^*}, p_C^{hu^*})$ such that no service provider has an incentive to deviate from their chosen price, after considering other service providers' prices, i.e.,*

$$W_I(p_I^{hu^*}; p_N^{hu^*}, p_C^{hu^*}) \ge W_I(p_I^{hu}; p_N^{hu^*}, p_C^{hu^*}), \forall p_I^{hu} \in [0, \infty);$$
$$W_{NC}(p_N^{hu^*}, p_C^{hu^*}; p_I^{hu^*}) \ge W_{NC}(p_N^{hu}, p_C^{hu}; p_I^{hu^*}),$$
$$\forall p_N^{hu} \in [0, \infty), p_C^{hu} \in [0, \infty).$$

We present service providers' optimal equilibrium prices in Theorem 4.

Theorem 4. *The optimal equilibrium prices of IoTSP, NSP, and CSP in the horizontally-uncoordinated structure are given in (23).*[12]

Proof of Theorem 4 is given in Appendix 6, available in the online supplemental material. Based on Theorem 4, we can show that the total equilibrium price for customers (i.e., $p_I^{hu^*} + p_N^{hu^*} + p_C^{hu^*}\xi$) increases in each service provider's cost. However, such a monotonic property does not necessarily hold for each of the prices.

Proposition 1. *In the horizontally-uncoordinated structure, given a fixed IoTSP's cost C_I, the optimal total equilibrium price of NSP and CSP (i.e., $p_N^{hu^*} + p_C^{hu^*}\xi$) increases in their own cost (i.e., $C_N + C_C\xi$). For IoTSP's optimal equilibrium price (i.e., $p_I^{hu^*}$), there are two cases:*

1) *if $d^{\max} \le 2\, d^{\min}$, $p_I^{hu^*}$ decreases in cost $C_N + C_C\xi$.*
2) *if $d^{\max} > 2\, d^{\min}$, $p_I^{hu^*}$ first decreases in resource cost $C_N + C_C\xi$ when $C \le \frac{d^{\min}}{d^{\max 2}} A$, then increases when $\frac{d^{\min}}{d^{\max 2}} A < C \le \frac{A}{4\, d^{\min}}$, and finally decreases when $\frac{A}{4\, d^{\min}} < C \le \frac{A}{d^{\min}}$.*

12. The specific values of $p_N^{hu^*}$ and $p_C^{hu^*}$ can be chosen arbitrarily, as long as their weighted sum equals the value in (19) and they are larger than the respective costs (i.e., C_N and C_C).

Proof of Proposition 1 is given in Appendix 7, available in the online supplemental material. As illustrated in Fig. 5, Proposition 1 shows that:

- In most cases, IoTSP's optimal equilibrium price decreases in other service providers' cost. This is because other service providers' price increases in their cost, so to maximize its profit, IoTSP needs to avoid a large customer demand reduction due to the price rise.
- The case where IoTSP's optimal price increases in other service providers' cost happens when $d^{\max}$ is large and cost $C_N + C_C\xi$ is medium, i.e., $C_N + C_C\xi \in [1, 2.5]$ of the red curve in Fig. 5b. In this range, as customer distribution is sparse (i.e., $d^{\max} > 2\, d^{\min}$) and most customers have already used the service (due to the medium cost $C_N + C_C\xi$), customers' demand is inelastic, so increasing the price increases the profit. As we shall seen in Proposition 3, the selfish price rise behaviors will lead to the inefficiency of the whole system.

Next we will compare the performances of the interaction structures to see which benefits the providers and the customers most.

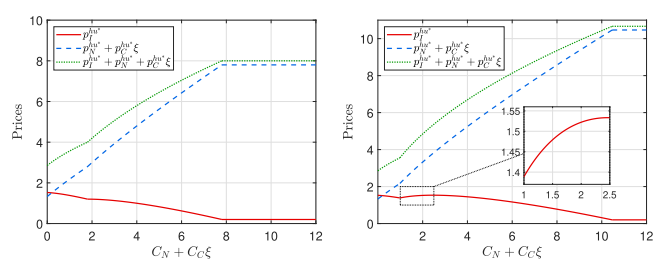
5 Comparison of Three Interaction Structures

In this section, we compare three interaction structures in terms of service providers' prices, profits, market share, as well as customers' payoffs. We are interested in understanding which structure is the best in each of four metrics.

In Section 5.1, we will compare the coordinated and the uncoordinated structures. In Section 5.2, we will further compare the two uncoordinated structures.

5.1 Coordinated versus Uncoordinated

We first try to understand whether coordination is better than uncoordination in terms the four metrics. For the convenience of presentation, we first specify several notations. We use p to denote the total unit price for customers, which



(a) $d^{\max} \le 2d^{\min}$. (b) $d^{\max} > 2d^{\min}$.

Fig. 5. Service providers' optimal equilibrium prices versus the cost of NSP and CSP in the horizontally-uncoordinated structure.

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equals p_I in coordinated and vertically-uncoordinated structures and equals $p_I + p_N + p_C\xi$ in the horizontally-uncoordinated structure. We denote the fraction of customers who use the IoT service as R (market share).

We present the comparison results in Proposition 2.

Proposition 2. *The optimal total unit price for customers in coordinated and uncoordinated structures satisfies*

$$\min\{p^{vu^*}, p^{hu^*}\} \geq p^{co^*}. \tag{24}$$

The total profit of service providers satisfies

$$\max\{W^{vu^*}, W^{hu^*}\} \leq W^{co^*}. \tag{25}$$

Each customer n's payoff satisfies

$$\max\{U_n^{vu^*}, U_n^{hu^*}\} \leq U_n^{co^*}, \forall n \in \mathcal{N}. \tag{26}$$

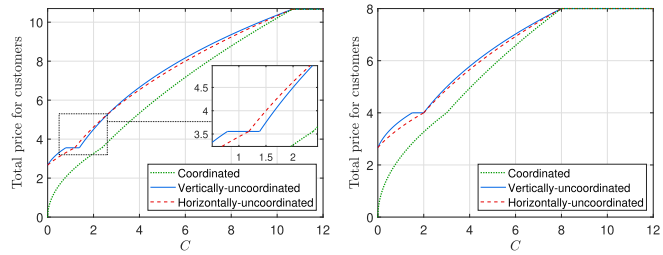
Service providers' market share satisfies

$$\max\{R^{vu^*}, R^{hu^*}\} \leq R^{co^*}. \tag{27}$$

Proof of Proposition 2 is given in Appendix 8, available in the online supplemental material. Proposition 2 demonstrates that both the service providers and the customers prefer the coordinated structure than the uncoordinated structures[13]. This is surprising at the first glance, as we would expect that the service providers' gain would be the customers' loss. We explain the reasons as follows.

- For service providers' total profit comparison in (25), it is because service providers in the coordinated structure jointly maximize their total profit, while they have diverse objectives and selfish behaviors in the uncoordinated structures.
- As shown in (24), service providers will set a higher total price for customers in the uncoordinated structures than the coordinated structure. First, for the vertically-uncoordinated structure, it is due to the *double marginalization effect* in economics [44], where different firms (service providers here) at different vertical levels apply their own markups in prices. Second, for the horizontally-uncoordinated structure, it is counter-intuitive as we may presume that the competition of service providers would lead to lower prices. This is because service providers in this case have complementary products, which means for each customer's usage, each provider's service is a necessary component. We name this phenomenon as *price-rise competition effect*, and similar phenomenon is also observed in other area's work (e.g., [45]).
- Due to the smaller price for customers in coordinated structure, customers will obtain higher payoffs (i.e., (26)), which leads to a larger fraction of participation (i.e., market share (27)).

13. Since service providers' total profit in coordinated structure is larger than uncoordinated structures, they can always find a profit distribution mechanism (e.g., Nash bargaining [43]) to make each provider obtain a larger profit in coordinated structure than uncoordinated structures.



(a) $d^{\max} > 2d^{\min}$. (b) $d^{\max} \leq 2d^{\min}$.

Fig. 6. Total price for customers versus total cost of service providers in three interaction structures.

5.2 Uncoordinated: Vertical Vs. Horizontal

Recall that service providers provide integrated services to customers in the vertically-uncoordinated structure, while customers need to purchase services from each service provider in the horizontally-uncoordinated structure. In this subsection, we will compare the two uncoordinated structures to see if the service integration benefits uncoordinated service providers or customers.

The comparison of the vertically and horizontally uncoordinated structures is given in Propositions 3.

Proposition 3. *There are two cases for the comparison of vertically-uncoordinated and horizontally-uncoordinated structures:*

- *Case 1. If $d^{\max} > 2\ d^{\min}$ and $4\ d^{\min} < \frac{A}{C} < \frac{d^{\max 2}}{d^{\min}}$, the vertically-uncoordinated structure is better than the horizontally-uncoordinated structure in terms of all four metrics, i.e.,*

$$p^{vu^*} < p^{hu^*}, \tag{28}$$
$$W^{vu^*} > W^{hu^*}, \tag{29}$$
$$U_n^{vu^*} \geq U_n^{hu^*}, \ \forall n \in \mathcal{N}, \tag{30}$$
$$R^{vu^*} > R^{hu^*}. \tag{31}$$

- *Case 2. Otherwise, the horizontally-uncoordinated structure is better than the vertically-uncoordinated structure in terms of four metrics.*

Proof of Proposition 3 is given in Appendix 9, available in the online supplemental material. Proposition 3 shows that when there are a large customer demand variance and a medium utility-cost ratio, the vertically-uncoordinated structure is more beneficial to both service providers and customers. As we have explained in Proposition 1, under the condition of Case 1 in Proposition 3, all service providers in the horizontally-uncoordinated structure will selfishly choose to increase their prices due to customers' inelastic demand, resulting in the inefficiency of the whole system. As validated in Fig. 6, only when $C \in [1.2, 2.7]$ in Fig. 6a (corresponding to Case 1), the horizontally-uncoordinated structure has a larger price for customers.

So far, we have analyzed the optimal pricing mechanism and interaction structures for a generic IoT system. Next, motivated by some specific scenarios (e.g., IoV) where each customer device has considerable computation capability, we are interested in extending the model by further considering customers' local computation.

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$$p^* = \begin{cases} \sqrt{\frac{2A(C^v+e)}{d^{\max}+d^{\min}}} - e, & \text{if } C^v \le \frac{d^{\max}+d^{\min}}{2\,d^{\max 2}} A - e, & (34a) \\ \frac{\sqrt{A^2+8A(C^v+e)d^{\min}}-A}{2\,d^{\min}} - e, & \text{if } \frac{d^{\max}+d^{\min}}{2\,d^{\max 2}} A - e < C^v < \frac{A}{d^{\min}} - e, & (34b) \\ \frac{A}{d^{\min}} - e, & \text{if } C^v \ge \frac{A}{d^{\min}} - e & (34c). \end{cases}$$

6 Model Extension: Partial Local Computation

In this section, motivated by the specific scenarios (such as IoV), we will extend our model by adding some features about customers' local computation and investigating their impacts on the whole system. Specifically, we first update the system model by considering the additional important feature of partial local computation of customers in Section 6.1, then calculate customers' optimal decisions and service providers' optimal prices in Section 6.2, and finally study the impacts of new features in Section 6.3.

As we have shown in Proposition 2, coordinated interaction structure (where IoTSP, NSP, and CSP cooperate with each other) is the best for both service providers and customers. Thus, we consider that service providers form a coordinated interaction structure in this section. Note that we also analyze the two uncoordinated structures through similar approaches in Section 4, and we verify that the structure comparison results in Section 5 are still true under this extension. Due to space limit, we present these results in Appendix 16, available in the online supplemental material, and we will validate the results through simulation in Section 7.3.

6.1 System Model Updates

As mentioned in Section 1.1, a current paradigm in some scenarios such as IoV is that customers locally process some data and upload the remaining data to cloud. In this case, the customers' computation costs consist of two parts:

- for the local data processing, customers have local computation costs (e.g., in energy or time);
- for the cloud data processing, customers have cloud computation costs (e.g., data privacy cost and transmission cost) and payment to NSP and CSP.

We denote the fraction of the computation load that each customer locally processes as k, the unit cloud computing cost as e^c,[14] and the unit local computation cost as e^l.[15]

In the coordinated case, a customer n's payoff is the difference between his utility and costs:

$$U_n = A\log(1+\ell_n) - [p_I + ke^l + (1-k)(p_N + p_C\xi + e^c)]d_n\ell_n, \tag{32}$$

where the unit cost is composed of three parts: the payment to IoTSP (i.e., p_I), the local computation cost (i.e., ke^l) and the cloud computation cost (i.e., $(1-k)(p_N + p_C\xi + e^c)$).

The service providers' joint profit in the coordinated structure is

$$W = \frac{1}{d^{\max} - d^{\min}} \cdot \int_{d^{\min}}^{d^{\max}} \ell d[(p_I - C_I) + (1-k)(p_N + p_C\xi - C_N - C_C\xi)]\mathrm{d}d, \tag{33}$$

where $(p_I - C_I)$ represents the unit IoT service profit, and $(1-k)(p_N + p_C\xi - C_N - C_C\xi)$ represents the unit profit form cloud computation.

6.2 Optimal Strategies of Customers and Providers

We still use the backward induction to analyze the customers' optimal decisions in Stage II and service providers' optimal prices in Stage I.

6.2.1 Stage II: Customers' Optimal Decisions

Given service providers' prices in Stages I, each customer n decides his optimal service usage level ℓ_n to maximize his payoff in Stage II by solving the following problem:

Problem 6 (Customer n's Optimization Problem).

$$\max\ A\log(1+\ell_n) - [p_I + ke^l + (1-k)(p_N + p_C\xi + e^c)]d_n\ell_n$$
$$\text{var. } \ell_n \in [0,\infty). \tag{35}$$

For the convenience of presenting the solutions of Problem 6, we introduce the following definitions:

$$p \triangleq p_I + (1-k)(p_N + p_C\xi), \tag{36}$$
$$e \triangleq ke^l + (1-k)e^c, \tag{37}$$

where term p represents total unit price for customers set by three service providers, and term e represents total unit computation cost, including both local and cloud computation costs.

Given these definitions, we present the closed form solutions of customers' optimal decisions as follows:

Lemma 3. *A customer n's optimal IoT service usage level in the coordinated structure is*

$$\ell_n^* = \begin{cases} \frac{A}{(p+e)d_n} - 1, & \text{if } p \le \frac{A}{d_n} - e, & (38a) \\ 0, & \text{if } p > \frac{A}{d_n} - e. & (38b) \end{cases}$$

Proof of Lemma 3 is given in Appendix 10, available in the online supplemental material. As shown in Lemma 3, besides the insights for Lemma 1 in the basic IoT scenario in Section 3, customers' optimal decisions in this extension are additionally affected by the computation costs (i.e., e^c and

14. Note that the term e^c is a customer's unit cloud computation cost (such as the customers' data privacy cost), which does not include the payment to CSP (i.e., p_C).

15. We consider that these three parameters are homogeneous for customers. It is convenient for the car operator to manage its vehicles with a uniform local computation fraction, and customers in the same system usually use the same devices and technologies. Moreover, our analysis approach can be easily applied to the case where these three parameters are all heterogeneous, and we will show through analysis and simulation in Section 7.4 that the key insights still remain valid.

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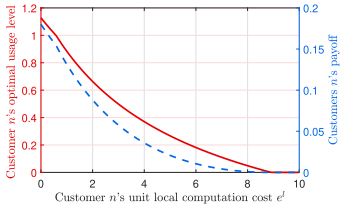

Fig. 7. Customer n's optimal usage level and payoff under different local computation costs e^l. ($d_n = 0.1, e^c = 0.5$, and $k = 0.6$.).

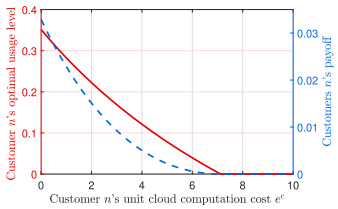

Fig. 8. Customer n's optimal usage level and payoff under different cloud computation costs e^c. ($d_n = 0.1, e^l = 4.5$, and $k = 0.6$.).

e^l) and local computation fraction k (as a part of p and e). After solving the optimal price p^* in Stage I, we will present the impacts of e^c, e^l, and k on customers' optimal strategies ℓ_n^* in Proposition 4 in Section 6.3.

6.2.2 Stage I: Service Providers' Optimal Price

Considering the optimal strategies of customers in Stage II, service providers decide the optimal price p^* to maximize its profit in Stage I.

The service providers' optimization problem is

Problem 7 (Service Providers' Optimization Problem).

$$\max \frac{1}{d^{\max} - d^{\min}} \cdot \int_{d^{\min}}^{d^{\max}} \ell d[(p_I - C_I) + (1-k)(p_N + p_C\xi - C_N - C_C\xi)]\mathrm{d}d,$$
$$\text{var. } p_I \in [0,\infty), p_N \in [0,\infty), p_C \in [0,\infty). \tag{39}$$

Similar to the analysis in Section 3.2, we obtain service providers' optimal price as follows:

Theorem 5. *The service providers' optimal price in the coordinated structure is given in (34), where $C^v \triangleq C_I + (1-k)(C_N + C_C\xi)$ is service providers' total unit cost.*

Proof of Theorem 5 is given in Appendix 11, available in the online supplemental material. As shown in Theorem 5, although the equation structure is similar to that in Theorem 1, the parameters e^c, e^l, and k affect the optimal price p^* in a complex way (through C^v and e). We will further analyze these impacts in next subsection.

6.3 Impacts of New Features

In this subsection, we will study the impacts of customers' unit cloud computation cost e^c, unit local computation cost e^l, and local computation fraction k.

6.3.1 Impacts on Customers' Optimal Decisions and Payoffs

After substituting the optimal price p^* into customers' optimal strategies ℓ_n^* and payoffs U_n^*, we can prove that

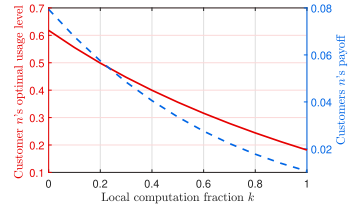

Fig. 9. Customer n's optimal usage level and payoff under different local computation fraction k when $e^l \geq e^c + C_N + C_C\xi$. ($C_N + C_C\xi = 1.75$, $d_n = 0.1, e^l = 4.5$, and $k = 0.6$.).

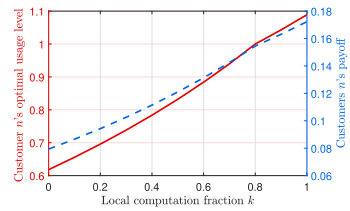

Fig. 10. Customer n's optimal usage level and payoff under different local computation fraction k when $e^l < e^c + C_N + C_C\xi$. ($C_N + C_C\xi = 1.75$, $d_n = 0.1, e^l = 1$, and $k = 0.6$.).

Proposition 4.

1) *Both customer n's optimal usage level ℓ_n^* and payoff U_n^* decrease in his computation costs e^c and e^l.*
2) *There are two cases for the impact of local computation fraction k on ℓ_n^* and U_n^*:*
   - *If $e^l \geq e^c + C_N + C_C\xi$, both ℓ_n^* and U_n^* decrease in k.*
   - *If $e^l < e^c + C_N + C_C\xi$, both ℓ_n^* and U_n^* increase in k.*

Proof of Proposition 4 is given in Appendix 12, available in the online supplemental material. The impact of k depends on the comparison of customers' local computation cost and cloud computation cost.[16] If the local computation cost is larger than the cloud computation cost (i.e., $e^l \geq e^c + C_N + C_C\xi$), then a larger local computation fraction k will lead to a higher cost for customers and thus decrease customers' usage levels and payoffs. Otherwise, a larger k will benefit customers. The corresponding illustration of Proposition 4 is shown in Figs. 7, 8, 9, and 10.

6.3.2 Impacts on Service Providers' Optimal Prices

We analyze how the optimal price changes with e^c, e^l and k in Propositions 5 and 6, respectively.

Proposition 5. *There are two cases for the impact of e^c and e^l on the optimal price p^*:*

- *if $d^{\max} \leq 2d^{\min}$, p^* first increases and then decreases in e^c and e^l. The maximum point is $e = \frac{A}{2(d^{\max}+d^{\min})} - C^v$.*
- *if $d^{\max} > 2\,d^{\min}$, p^* first increases in $e \in [0, \frac{A}{2(d^{\max}+d^{\min})} - C^v]$, decreases in $e \in [\frac{A}{2(d^{\max}+d^{\min})} - C^v, \frac{d^{\max}+d^{\min}}{2\,d^{\max 2}}A - C^v]$, increases in $e \in [\frac{d^{\max}+d^{\min}}{2\,d^{\max 2}}A - C^v, \frac{3}{8}\frac{A}{d^{\min}} - C^v]$, and decreases in $e \in (\frac{3}{8}\frac{A}{d^{\min}} - C^v, +\infty)$.*

16. Note that the $C_N + C_C\xi$ corresponds to the payment to NSP and CSP, which is a part of customers' cloud computation.

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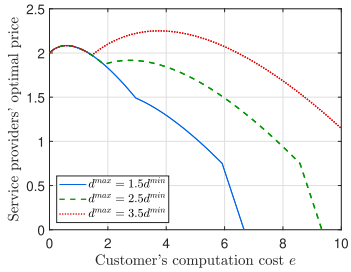

Fig. 11. Impact of customers' computation cost e on service providers' optimal price p^*.

- *The maximum point is* $e = \frac{A}{2(d^{\max}+d^{\min})} - C^v$ *if* $2\ d^{\min} < d^{\max} < 3\ d^{\min}$.
- *The maximum point is* $e = \frac{3}{8}\frac{A}{d^{\min}} - C^v$ *if* $d^{\max} \geq 3\ d^{\min}$.

Proof of Proposition 5 is given in Appendix 13, available in the online supplemental material. Proposition 5 shows that the maximum point of the price always happens at a medium cost e. When computation costs e^c and e^l are small, service providers will set a small price for customers, as customers' demand in this case is elastic. When e^c and e^l are large, service providers will also set a small price for customers to avoid customers' poor participation.

Moreover, as illustrated in Fig. 11, customers' distribution variance can greatly affect the impact of e^c and e^l. When the customers distribution variance increase, the maximum point also becomes larger (or at least remains unchanged), i.e., the highest price happens at a higher cost e. Also, for the same cost e, service providers tend to set a higher price when distribution variance is larger. These observations are due to fact that when customers' distribution variance is larger, there will be more customers who have smaller resource demand d_n and thus can afford higher price.

Proposition 6. *The impact of local computation fraction k on the optimal price p^* depends on the costs and demand distribution, which is summarized in Table 2.*

Proof of Proposition 6 is given in Appendix 14, available in the online supplemental material. The local computation fraction k affects the optimal price in a complex way, and we give a complete characterization of the complex impact cases in Table 2.

- For each line in Table 2, we read from left to right. The $\uparrow$ represents p^* increases in k and $\downarrow$ represents p^* decreases in k. For example, in the first line, p^* first increases, then decreases, then increases, then decreases, and finally decreases in k.
- Three shadow regions: blue means full customer participation (corresponding to Equation (34a)), green means partial participation (Equation (34b)), and pink means no participation Equation (34c).
- When customers' local computation cost e^l is large (i.e., top half of Table 2), increasing k makes the optimal price move from blue region, to green region, and to pink region, which means that customers are less likely to participate; when e^l is small (i.e., bottom half of Table 2), increasing k makes customers more willing to participate. These are consistent with the results in Proposition 4.

Note that as k takes the value from 0 to 1, sometimes we need to take this segment from above cases. For example, under the first condition in Table 2 (i.e., $e^l \geq e^c + 2(C_N + C_C\xi)$ and $d^{\max} > \frac{2(e^l-e^c)}{e^l-e^c-C_N-C_C\xi} d^{\min}$),

- As illustrated in Fig. 12a, when $k = 0$ corresponds to the blue region and $k = 1$ corresponds to the red region (i.e., in mathematics $C_I + C_C\xi + C_N + e^c < (d^{\max} + d^{\min})A/(2\ d^{\max^2})$ and $C_I + e^l > A/d^{\min}$), the curve covers three regions (i.e., blue, green, and pink);
- As shown in Fig. 12b, when $k = 0$ corresponds to the green region and $k = 1$ corresponds to the red region (i.e., $(d^{\max} + d^{\min})A/(2\ d^{\max^2}) < C_I + C_C\xi + C_N + e^c < A/d^{\min} < C_I + e^l$), the curve only takes a segment in the green and red regions;
- As an example in Fig. 12c, when both $k = 0$ and $k = 1$ correspond to the blue region (i.e., $C_I + e^l < (d^{\max} + d^{\min})A/(2\ d^{\max^2})$), the curve only takes a segment in the blue region.

In summary, the impact of k on providers' optimal price mainly depends on customers' various costs comparison and distribution variance, which fundamentally determine customers' demands, payoffs, and participation decisions.

TABLE 2
Impact of Local Computation Fraction k on Optimal Price p^*

Conditions on the Costs and Demand Distribution		Relationship Between p^* and k				
$e^l \geq e^c + 2(C_N + C_C\xi)$	$d^{\max} > \frac{2(e^l-e^c)}{e^l-e^c-C_N-C_C\xi} d^{\min}$	$\nearrow$	$\searrow$	$\nearrow$	$\searrow$	$\searrow$
	$d^{\max} \leq \frac{2(e^l-e^c)}{e^l-e^c-C_N-C_C\xi} d^{\min}$	$\nearrow$	$\searrow$	$\searrow$		$\searrow$
$e^c + C_N + C_C\xi < e^l < e^c + 2(C_N + C_C\xi)$		$\nearrow$	$\searrow$	$\searrow$		$\searrow$
$e^c \leq e^l \leq e^c + C_N + C_C\xi$		$\searrow$		$\searrow$		$\searrow$
$e^l < e^c$	$d^{\max} > \frac{2(e^l-e^c)}{e^l-e^c-C_N-C_C\xi} d^{\min}$	$\nearrow$		$\nearrow$	$\searrow$	$\nearrow$ $\searrow$
	$d^{\max} \leq \frac{2(e^l-e^c)}{e^l-e^c-C_N-C_C\xi} d^{\min}$	$\nearrow$		$\searrow$		$\nearrow$ $\searrow$

[a] $\uparrow$ represents p^* increases in k and $\downarrow$ represents p^* decreases in k. For example, under the first condition, p^* first increases, then decreases, then increases, then decreases, and finally decreases in k.

[b] Three shadow regions: blue means full customer participation in Equation (34a), green means partial participation in Equation (34b), and pink means no participation in Equation (34c).

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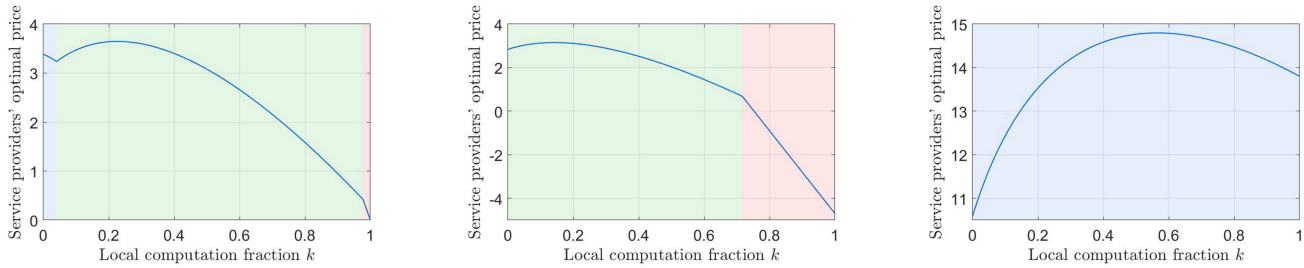

(a) Example of three regions. (b) Example of two regions. (c) Example of one region.

Fig. 12. When $e^l \geq e^c + 2(C_N + C_C\xi)$ and $d^{\max} > \frac{2(e^l-e^c)}{e^l-e^c-C_N-C_C\xi} d^{\min}$, different segments for the relationship between local computation fraction $k \in [0,1]$ and optimal price p^*.

6.3.3 Impacts on Service Providers' Profits

Although the optimal price has a complicated relationship with k, as we shall see in the following, the impact of k on service providers' profit is straightforward, i.e., the service providers can easily know how to adjust k to maximize their profit.

We present how service providers' profit changes with the parameters e^c, e^l and k in Proposition 7.

Proposition 7.

1) *Service providers' profit W^* (under the optimal strategies p^* and ℓ_n^*) decreases in e^c and e^l.*
2) *There are two cases of the impact of k on service providers' profit W^*:*
   - *If $e^l \geq e^c + C_N + C_C\xi$, W^* decreases in k.*
   - *If $e^l < e^c + C_N + C_C\xi$, W^* increases in k.*

Proof of Proposition 7 is given in Appendix 15, available in the online supplemental material. For the impact of computation costs (i.e., e^c and e^l), Propositions 4 and 7 show that decreasing customers' computation costs (either local or cloud) increases both customers' payoffs and service providers' profit. For example, service providers can decrease customers' cloud computation costs by improving confidentiality agreement and developing low-cost transmission technology, and decrease customers' local computation costs by designing appropriate processors in vehicles.

The impact of local computation fraction k depends on the comparison between local and cloud computation costs. Propositions 4 and 7 show that if customers' local computation cost is larger than the cloud computation costs, service providers should set a smaller k (e.g., through algorithms in vehicles), which will also give customers higher payoffs.

Next, we will evaluate the mechanism performance and present more insights through simulations.

7 Simulation Results

In this section, we conduct numerical experiments to evaluate the performance of our proposed pricing mechanisms. In Section 7.1, we compare our mechanisms with the state-of-the-art benchmarks. In Section 7.2, we further investigate the interaction structure superiority for each individual service provider. In Section 7.3, we compare the performance of three interaction structures under the extended model. In Section 7.4, we show that our key insights still remain valid under different demand distribution and multi-dimensional heterogeneity of customers.

We assume that customers' demands are uniformly distributed on $[d^{\min}, d^{\max}] = [0.1, 0.2]$, their utility coefficient is $A = 0.8$, their unit local computation cost is $e^l = 4.5$, their unit cloud computation cost is $e^c = 0.5$, and service providers' costs are $C_I = C_N + C_C\xi = 1.75$.[17]

7.1 Performance Evaluation

To evaluate the performance of our pricing mechanism, we list two benchmarks and our proposed mechanism as follows.

- *Homogeneous Demand Pricing (HD)* [30]: service providers design their optimal prices by assuming that customers have the same resource demand for each IoT service usage (i.e., $d_n = \bar{d}, \forall n \in \mathcal{N}$)[18].
- *Full Participation Pricing (FP)* [46]: service providers optimize their prices subject to the constraint that all customers can afford the services.[19]
- *Our proposed pricing mechanism (FH):* service providers Flexibly choose target customers and take customers' Heterogeneity into consideration.[20]

Fig. 13 illustrates service providers' total profit (W), IoTSP's profit (W_I), and the profit of NSP and CSP ($W - W_I$) in three interaction structures. Each service provider achieves a 21.00%-135.48% higher profit under our proposed FH pricing mechanism than the HD and FP benchmarks in each interaction structure.[21] Note that under FP, service providers may even choose not to participate in both uncoordinated structures, as they may get negative profits.

Fig. 14 shows that our proposed FH outperforms HD and FP in terms of social welfare (denoted by S, which is the sum of service providers' profits and customers' payoffs) by up to 122.55%, despite a possible smaller (at most 25.71% decrease) customers' total payoff (denoted by U). Customers' payoff can be higher in HD and FP, because service providers may inappropriately set lower prices for customers at the expense of providers' own profits.

17. This parameter setting is just one example for illustration. Although the numerical results/values will be parameter dependent, the key insights in this paper always hold for other parameter values.
18. We set $\bar{d} = (d^{\max} + d^{\min})/2$ in our simulations.
19. Due to space limit, we present the closed-form optimal prices in these two benchmark cases in Appendix 17, available in the online supplemental material.
20. As the benchmarks in literature do not consider partial local computation, we use the results derived in Sections 3 and 4 for comparison, instead of the extended results in Section 6.
21. We use Nash bargaining to distribute the profit among service providers in the coordinated structure.

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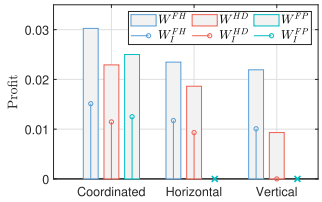


Fig. 13. Service providers' profit comparison of FH, HD, and FP in three interaction structures.

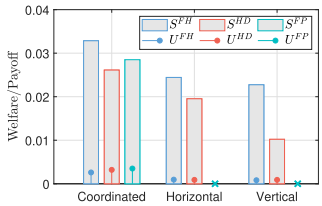


Fig. 14. Social welfare (S) and customers' total payoff (U) comparison of FH, HD, FP in three interaction structures.

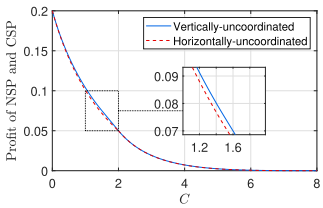


Fig. 15. Profit comparison of NSP and CSP in the vertically-uncoordinated and horizontally-uncoordinated structures. $d^{\max} = 0.225$, $d^{\min} = 0.075$.

7.2 Structure Comparison for Individual Service Providers

We have the following observations of each service provider's profit comparison in two uncoordinated structures:

- Fig. 15 shows that NSP and CSP obtain almost the same profits in two uncoordinated structures, with the difference less than 3.13%.
- Fig. 17 shows that only when customers' demand variance is large (i.e., $d^{\max} > 2\ d^{\min}$) and providers' cost is medium (i.e., $C \in [1.2, 2.7]$), IoTSP obtains a larger profit in the vertically-uncoordinated structure, due to the price-rise competition with NSP and CSP in the horizontally-uncoordinated structure (as we have explained in Section 5.2). Otherwise, IoTSP prefers the horizontally-uncoordinated structure, because in the vertically-uncoordinated structure, the price markup of NSP and CSP at Stage I harms IoTSP's interests in Stage II.

7.3 Structure Comparison Under the Extended Model

We compare service providers' profit under different values of $d^{\max}$ and d^{min} and different local computation fraction k in three interaction structures for the extended model.

In Fig. 16, we study service providers' total profit under different values of $d^{\max}$ and $d^{\min}$. Specifically, we keep the average demand fixed (i.e., $(d^{\max} + d^{\min})/2 = 0.15$) and study the cases of $d^{\max} = d^{\min}$, $d^{\max} = 2\ d^{\min}$, $d^{\max} = 3\ d^{\min}$, $d^{\max} = 4\ d^{\min}$, and $d^{\max} = 5\ d^{\min}$. The profits comparison

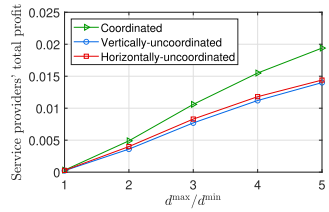
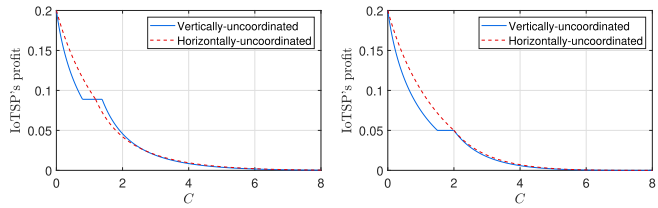
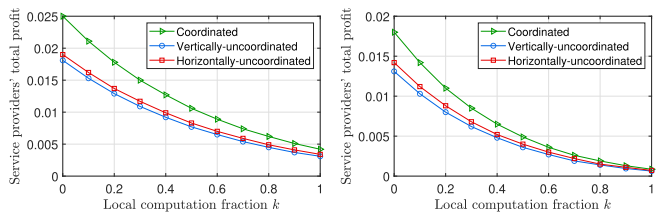


Fig. 16. Service providers' profit comparison under different values of $d^{\max}$ and d^{min} with three interaction structures.



(a) $d^{\max} > 2d^{\min}$. (b) $d^{\max} \leq 2d^{\min}$.

Fig. 17. Profit comparison of IoTSP in the vertically-uncoordinated and horizontally-uncoordinated structures. (a): $d^{\max} = 0.225, d^{\min} = 0.075$. (b): $d^{\min} = 0.1, d^{\max} = 0.2$.



(a) $d^{\max} > 2d^{\min}$. (b) $d^{\max} \leq 2d^{\min}$.

Fig. 18. Service providers' profit comparison under different local computation fraction k in three interaction structures.

with different interaction structures validate our structure comparison results for the extended model. Moreover, the results show that a larger demand variance will increase service providers' total profit.

Fig. 18 also shows that the structure comparison results (i.e., Propositions 2 and 3) are still true for the extended model. Specifically, coordinated structure makes service providers obtain a larger profit than the two uncoordinated structures; horizontally-uncoordinated is better than the vertical one when customers' distribution variance is small (i.e., $d^{\max} \leq 2\ d^{\min}$). Moreover, Fig. 18 also validates the impact of local computation fraction k on service providers' profit (i.e., Proposition 7). Under the experiment setting where the local computation cost is larger than cloud costs, increasing the local computation fraction will decrease service providers' profit.

7.4 Insights Validation Under Different Demand Distribution and Multi-Dimensional Heterogeneity

Next, we will show through analysis and simulations that some key insights still remain valid under different demand distribution of $g(d)$ and multi-dimensional heterogeneity of customers. The customer heterogeneity includes heterogeneous utility coefficients A_n, local computation fractions k_n, local computation costs e_n^l, and the

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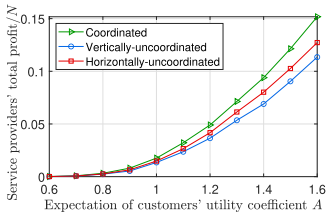

Fig. 19. Service providers' total profit under different expectations of customers' utility coefficients.

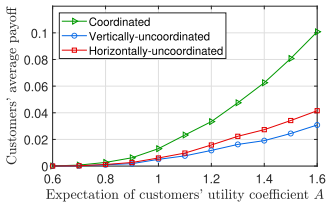

Fig. 20. Customers' average payoff under different expectations of customers' utility coefficients.

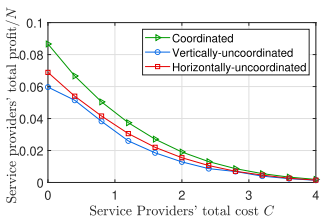

Fig. 21. Service providers' total profit under different total costs ($C = C_I + C_N + C_C\xi$) when $d^{\max} = 1.5$ $d^{\min} = 0.18$.

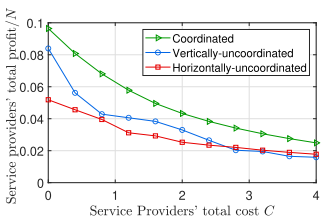

Fig. 22. Service providers' total profit under different total costs ($C = C_I + C_N + C_C\xi$) when $d^{\max} = 9$ $d^{\min} = 0.27$.

resource demand d_n. Due to space limit, we put the model updates and theoretical analysis (similar to this paper) in Appendix 18, available in the online supplemental material, and only show the simulation results in this section.

For the experiment setting, we consider $N = 5 \times 10^5$ customers. Customers' demands $\{d_n\}_{n\in\mathcal{N}}$ follow a truncated normal distribution $N(0.15, 0.03^2)$ over the support $[d^{\min} = 0.075, d^{\max} = 0.225]$, customers' utility coefficients $\{A_n\}_{n\in\mathcal{N}}$ follow a truncated normal distribution $N(0.8, 0.1)$ over the support $[0, +\infty)$, customers' local computation fractions $\{k_n\}_{n\in\mathcal{N}}$ follow a truncated normal distribution $N(0.5, 0.06)$ over the support [0,1], and customers' unit local computation cost $\{e_n^l\}_{n\in\mathcal{N}}$ follow a truncated normal distribution $N(4.5, 1)$ over the support $[0, +\infty)$. Customers' unit cloud computation cost $e^c = 0.5$, and service providers' costs $C_I = C_N + C_C\xi = 1.75$.

As we can see from Figs. 19 and 20, the coordinated structure is better than two uncoordinated structures for both service providers and customers, which is consistent

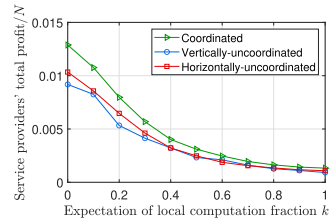

Fig. 23. Service providers' total profit under different expectations of customers' local computation fractions. Here $e^c = 0.5$.

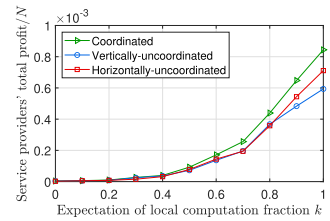

Fig. 24. Service providers' total profit under different expectations of customers' local computation fractions. Here $e^c = 3.5$.

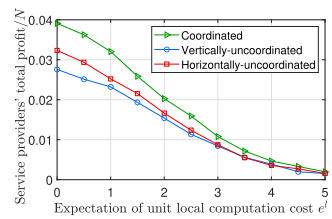

Fig. 25. Service providers' total profit under different expectations of customers' unit local computation costs.

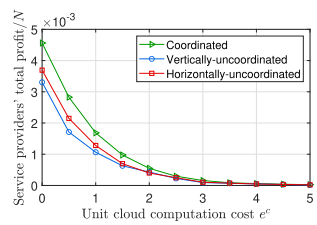

Fig. 26. Service providers' total profit under customers' different unit cloud computation costs.

with Proposition 2. Moreover, larger utility coefficients will increase both service providers and customers' profits or payoffs, as customers can afford higher costs and are more willing to participate. This has been indicated in Lemma 3 and Theorem 5.

Figs. 21 and 22 show that although the coordinated structure is always better than the two uncoordinated structures, the comparison of two uncoordinated structures depends on the customers' demand variance and service providers' costs. If customers' demand variance is small (as shown in Fig. 21), horizontally-uncoordinated structure is better than the vertically-uncoordinated structure. If the customers' demand variance is large (as shown in Fig. 22), it is possible that vertically-uncoordinated structure is better than the horizontally-uncoordinated structure. The insights are similar to Proposition 3.

Figs. 23 and 24 show the impacts of parameter k. Similar to the results in Proposition 7, as shown in Fig. 23, when the cloud computation cost is smaller than the local

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computation cost, service providers' profit decreases in the local computation fraction. As shown in Fig. 24, when the cloud computation cost is larger than the local computation cost, service providers' profit increases in the local computation fraction.

Figs. 25 and 26 show the impacts of parameters e^l and e^c. The results show that service providers' profit decreases in the local and cloud computation costs, which is consistent with Proposition 7.

In conclusion, the key insights of the paper remain the same when we further considering a different demand distribution (instead of uniform distribution) and multi-dimensional heterogeneity of customers.

8 Conclusion

To the best of our knowledge, this is one of the first papers to study service providers' optimal pricing and interaction structures in the presence of heterogeneous customers. We showed that coordination of service providers is better than uncoordination for both providers and customers, as it avoids uncoordinated providers' selfish price markup behaviors. When customers' demand variance is large and utility-cost ratio is medium, the vertically-uncoordinated structure is better than the horizontal one for both providers and customers because of complementary providers' competition in the horizontally-uncoordinated structure. We also showed that we can easily extend our generic model and obtain the impact of local computation. The simulations further showed that our proposed pricing mechanisms increase service providers' profits by up to 135.48% and social welfare by up to 122.55%, compared with the state-of-the-art benchmarks.

In the future work, we will use real-world data to validate our results and develop a more complicated and practical customer utility model for further study.

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